

Bootstrap Inference for Mutual Fund Alphas: Luck, Skill, and the Right Tail

Dhruv Kohli

University of Chicago Booth School of Business

Abstract

This paper asks whether the right tail of mutual fund alpha t -statistics is stronger than luck would produce in a large cross-section. The empirical design combines the Carhart four-factor model with the luck-versus-skill bootstrap logic of [Kosowski et al. \(2006\)](#). The WRDS-derived mutual-fund panel covers January 1987 through December 2025, or 468 monthly observations, with 9,658 fund columns and 6,764 funds in the audited baseline test after cleaning and the 60-month observation rule. The data screen flags economically implausible monthly returns, excludes those gated observations from every reported inferential run, and records the limitation that exact fund identifiers are not bundled with the analysis matrix. The evidence is conservative but direct. The audited sample contains impressive individual winners, yet the observed maximum and upper-tail alpha statistics sit well inside the joint zero-alpha bootstrap distribution. Under the Carhart benchmark and the implemented bootstrap design, the apparent winners are statistically indistinguishable from luck. Wild-bootstrap, block-bootstrap, covariance, filter-sensitivity, factor-sensitivity, and subperiod checks do not overturn that conclusion. The paper therefore replaces a fund-level winner narrative with a transparent cross-sectional test of whether apparent mutual-fund stars survive the luck benchmark.

1 Introduction

Active mutual fund performance looks like a skill question, but statistically it starts as a luck question. In a universe with thousands of funds, some managers will have large estimated alphas even if every true benchmark-adjusted alpha is zero. A single fund with a t -statistic above two is therefore not enough. The relevant object is the right tail of the full cross-section: the maximum, the 99th percentile, and nearby order statistics.

This paper asks whether that right tail is too large to be explained by chance under a Carhart four-factor benchmark. The answer is intentionally model-dependent. Alpha here means performance left over after exposure to the market, size, value, and momentum factors has been removed. It is not a model-free measure of talent, and it is not a claim about every possible benchmark.

The empirical scale is large enough for the multiple-testing problem to be first order. The mutual-fund panel is derived from WRDS and runs from January 1987 through December 2025, giving 468 monthly observations. It contains 9,658 mutual-fund columns, of which 6,764 enter the baseline after the outlier screen and the requirement of at least 60 valid monthly returns.

The cross-sectional units are funds, not operating firms, and the central statistical problem is how large the best fund statistics should look in a sample of that size.

The analysis also begins with a data-quality problem. The WRDS-derived return matrix contains monthly excess returns that are mechanically impossible or economically implausible, including values far above any credible mutual fund return. If those observations are left in the sample, data errors can be converted mechanically into evidence of skill. The first contribution is therefore not a new factor model; it is a disciplined sample construction step. Threshold-flagged extreme returns are excluded from every reported inferential run, and the paper states the documentation boundary created by the absence of fund identifiers in the analysis matrix.

The empirical design then moves in stages. First, the return and factor matrices are audited so the sample is explicit. Second, fund-level Carhart regressions describe the observed alpha- t distribution. Third, residual and wild bootstraps impose the joint zero-alpha null and test whether the observed right tail is unusual relative to that null. Fourth, robustness checks vary covariance estimators, degrees-of-freedom rules, factor specifications, fund-history filters, dependence assumptions, and subperiod windows.

The main result is direct. In this audited WRDS-derived sample, the observed right tail is not unusually strong once it is compared with the bootstrap distribution of the best funds that luck alone can generate. The conclusion is not that skill can never exist. It is that this evidence does not support a robust right-tail mutual-fund alpha claim after data cleaning, benchmark adjustment, and joint-null inference.

2 Literature and Methodological Positioning

The paper sits at the intersection of mutual-fund performance evaluation, multiple testing, and bootstrap inference. The factor-model layer follows [Fama and French \(1993\)](#) and [Carhart \(1997\)](#). Their models give a disciplined way to define abnormal performance, but they also create the joint-hypothesis problem. A positive intercept may be skill, but it may also be compensation for omitted risks or an artifact of the benchmark.

[Kosowski et al. \(2006\)](#) turn that benchmark-adjusted performance problem into a luck-versus-skill design. The key insight is that fund selection is an order-statistic problem. Investors and researchers do not look at a randomly chosen fund; they look at the best funds after the fact. The appropriate null must therefore show how large the best alpha t -statistics can become when the true alphas are zero. This paper keeps that logic while emphasizing sample quality, explicit null construction, and sensitivity to modeling choices.

The bootstrap layer follows the principle that the resampling world should impose the null being tested. For fund i , the pseudo-return should preserve factor exposure and residual variation while setting $\alpha_i = 0$. Studentizing the alpha estimate makes the statistic more comparable across funds with different histories and residual variances ([Hall, 1992](#); [Horowitz, 2001](#); [Davison and Hinkley, 1997](#)). Wild bootstrap methods provide heteroskedasticity-oriented checks ([Wu, 1986](#); [Mammen, 1993](#)), while block methods provide a simple sensitivity test for serial dependence ([Künsch, 1989](#)). Because financial panels are dependent and unbalanced, the bootstrap method and the statistic have to be interpreted together rather than treated as interchangeable

(Gonçalves and Vogelsang, 2006).

This literature sets the language for the rest of the paper. The empirical question is not whether any fund can ever have skill. It is whether the audited matrix contains a right tail that is too strong for the selected benchmark and bootstrap null to explain. That distinction keeps the empirical claim useful without overstating what the design can identify.

3 Data and Sample Construction

The empirical input is a WRDS-derived mutual-fund return panel matched to monthly Carhart factors. The sample is not simulated and is not manually constructed. The data screen has two purposes: to state the period and cross-sectional coverage clearly, and to prevent extreme observations from mechanically determining right-tail inference.

The monthly panel runs from January 1987 through December 2025. That is 468 months, or 39 calendar years of data. The WRDS-derived return matrix has 9,658 fund columns. Because funds enter and exit the sample at different times, the effective baseline cross-section is smaller: after cleaning and requiring at least 60 valid monthly excess returns, 6,764 funds remain for the main bootstrap test.

The first empirical task is therefore a sample audit, not a regression. The screen checks dimensions, finite values, missingness, factor ranges, sample dates, and return magnitudes. The exclusion rule is deliberately asymmetric: monthly excess returns below -120% or above 100% are excluded from the clean analysis matrix. A roughly -100% monthly loss is not automatically removed, because a near-total fund loss can be economically possible in a levered or liquidating fund. In this panel, no return falls below the -120% lower bound; all 74 excluded observations are positive returns above 100% . Those gated observations are not used in any reported inferential run because they can mechanically dominate the right tail and are plausible data errors.

Table 1: WRDS-derived panel audit. The primary clean matrix excludes threshold-flagged extreme monthly returns.

Audit item	Value
Sample period	1987-01–2025-12
Monthly observations	468
Fund columns	9,658
Observations with $ r > 100\%$	75
Flagged below -120%	0
Flagged above 100%	74
Excluded primary-matrix observations	74
Excluded-observation funds	45
Most extreme excluded return	12012.415
Clean funds with ≥ 60 months	6,764
Fund identifiers recovered	No

Table 1 fixes the sample used in the rest of the analysis: January 1987–December 2025, 468 monthly rows, 9,658 fund columns, and 6,764 baseline funds after cleaning and the 60-month

rule. The table also shows why the screen is necessary. The raw matrix contains 75 observations with absolute monthly returns above 100%, but the exclusion rule drops 74 observations: zero below -120% and 74 above 100% . The remaining absolute-value case is a roughly -100% excess return, which stays in the clean matrix because it does not cross the lower-bound screen. The excluded positive returns are spread across 45 fund columns and 46 calendar months, with the largest value equal to 12,012.4 in monthly excess-return units. This matters because the right tail of alpha statistics is exactly where a small number of bad observations can do the most damage.

The analysis matrix also has a documentation limit. Exact `crsp_fundno` and `crsp_portno` identifiers are not bundled with the return columns. The month sequence is recovered from the date range, but the fund columns are unlabeled in the analysis array. Consequently, the paper treats the file as a WRDS CRSP mutual-fund extract while being precise about what is contained in the research inputs:

WRDS-derived mutual-fund returns are the empirical input; fund identifiers are not bundled with the analysis matrix

This limitation defines the documentation boundary of the analysis matrix. The matrix-level bootstrap does not require fund names, because the inferential object is the cross-sectional right tail. Any future fund-by-fund interpretation, however, would require identifiers alongside the return columns.

4 Bootstrap Design and Null Construction

The econometric design has one purpose: compare the best observed funds with the best funds that would appear under a zero-alpha world. For fund i and month t , the baseline return model is

$$R_{i,t} = \alpha_i + \beta_i^\top F_t + \varepsilon_{i,t}, \quad (1)$$

where $R_{i,t}$ is monthly excess return and F_t contains market, size, value, and momentum factors. A fund enters the baseline cross-section when it has at least 60 nonmissing monthly observations in the clean matrix.

For each fund, the ordinary least squares estimator is

$$\hat{b}_i = (X_i^\top X_i)^{-1} X_i^\top y_i, \quad (2)$$

and the baseline studentized alpha statistic is

$$t_i = \hat{\alpha}_i / \widehat{\text{se}}(\hat{\alpha}_i). \quad (3)$$

OLS standard errors are the baseline because they match the classical mutual-fund bootstrap comparison. That choice is not a claim that OLS errors are perfect for monthly fund returns. To study whether the observed tail is mainly a studentization artifact, the robustness section recomputes the same order statistics with HC3 and Newey–West standard errors.

The joint null is

$$H_0 : \alpha_i = 0 \quad \text{for every fund } i. \quad (4)$$

This null is stronger and more relevant than asking whether one fund is significant in isolation. The test statistics are order statistics of the cross-sectional t -statistic distribution: the maximum and selected upper percentiles. To verify whether the observed top funds are unusual, each bootstrap draw creates a new no-alpha panel, re-estimates every fund, and records the same order statistics. Bootstrap p -values use the plus-one correction,

$$p_j = \frac{1 + \#\{b : T_{j,b}^* \geq T_j^{obs}\}}{B + 1}, \quad (5)$$

and Monte Carlo standard errors are reported as $\sqrt{p_j(1 - p_j)/(B + 1)}$.

The primary cross-sectional residual bootstrap is designed to keep the calendar panel structure visible. It estimates the factor model, recenters residuals fund-by-fund, draws common month indices with replacement, pairs those sampled factor rows with sampled residual rows, constructs pseudo-returns under $\alpha_i = 0$, and re-estimates all fund regressions. Common month draws are used because mutual funds share market months. A draw that resamples funds independently would break the main common-shock structure that the cross-section actually experienced.

The baseline degrees of freedom are the number of valid sampled observations minus the number of estimated parameters. The older unique-month degrees-of-freedom adjustment is kept only as a robustness column because it changes the statistic in a nonstandard way. That column answers a sensitivity question; it is not the main specification.

The analysis then varies the main researcher choices. To verify whether the 60-month fund filter drives the conclusion, the same cross-sectional residual bootstrap is rerun with 36-, 60-, and 120-month entry rules. To verify whether apparent alpha is compensation for omitted factor exposure, the baseline fund universe is held fixed and the factor set is changed. This is important: changing the factor model while also changing the fund universe would mix two mechanisms. The factor-sensitivity exercise therefore isolates benchmark choice from sample-selection choice.

The wild bootstrap asks a different question. Instead of resampling calendar months, it multiplies recentered residuals by random weights and then rebuilds no-alpha pseudo-returns. The implementation uses the standard two-point Mammen distribution, with Rademacher and time-clustered multiplier variants as robustness. The full-clean-sample wild run uses every audited fund with enough observations to estimate a regression. This test asks whether the 60-month baseline is hiding evidence by excluding short-lived funds. Independent fund-time wild weights are useful for heteroskedasticity-oriented checks, but they do not preserve residual cross-correlation across funds. The time-clustered multiplier variant is included to study common calendar shocks more directly.

Finally, the subperiod analysis is deliberately labeled exploratory. The data begin in 1987, so a literal pre-law test around major anti-insider-trading statutes is not credible: the available pre-window is too short for a four-factor bootstrap. Instead, the paper studies within-sample information and crisis windows, including pre- and post-Reg FD windows and pre- and post-

GFC windows. These tests are reported with Holm adjustments and Monte Carlo-error flags. A window only supports a strong period-specific statement if the raw and adjusted evidence reject the joint null and the result is not borderline.

5 Empirical Results

Observed Alpha Distribution

The first result is deliberately descriptive. Before asking whether the best funds are unusually good, the baseline factor regressions show what the observed alpha distribution looks like. Table 2 reports the baseline cross-section after the 60-month filter. The average alpha is negative, the mean and median t -statistics are both below zero, and yet the maximum t -statistic is large. That combination is exactly why the paper uses order-statistic inference: the average fund looks weak, but the best observed fund can still look impressive in a large universe.

Table 2: Fund-level four-factor OLS summary on the audited primary matrix.

Statistic	Value
Baseline funds	6,764
Valid OLS t -statistics	6,764
Mean alpha (% monthly)	-0.144
Mean alpha (% annualized)	-1.73
Mean t -statistic	-1.095
Median t -statistic	-1.125
Max t -statistic	6.350
Min t -statistic	-8.913
Funds with $t > 2$	76
Funds with $ t > 1.96$	1,739

Figure 1 shows the same point visually. The distribution is centered to the left of zero, but both tails are wide. The right tail contains funds that would be tempting to describe as skilled if viewed one at a time. The left tail is a useful warning: extreme underperformance is also common in a large noisy panel, so the paper should not turn tail observations into a skill story without a joint-null benchmark.

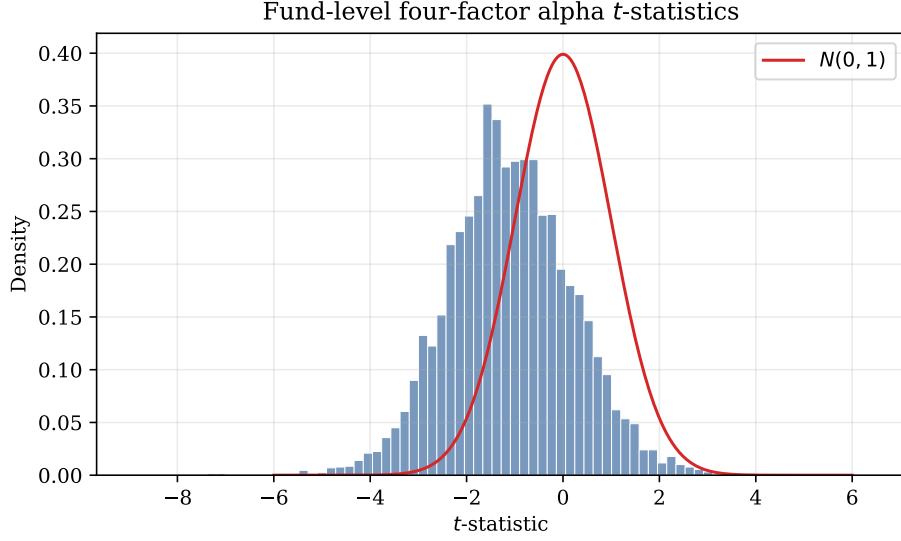


Figure 1: Distribution of fund-level alpha t -statistics in the audited baseline sample.

Primary Joint-Null Test

The main test asks whether those right-tail statistics exceed what the no-alpha world can generate. To verify this, Table 3 compares the observed maximum and upper percentiles with the configured cross-sectional residual-bootstrap draws. Each draw preserves the factor model structure, imposes zero alpha, resamples common calendar months, and then re-estimates every fund. The resulting p -value is therefore a statement about the best funds in the cross-section, not about a randomly selected fund.

Table 3: Primary cross-sectional residual bootstrap.

Statistic	Observed t	p -value	MC s.e.
max	6.350	0.8780	0.0511
99.5%	2.428	0.9024	0.0463
99%	2.108	0.9024	0.0463
98%	1.606	0.9512	0.0336
97%	1.388	0.9512	0.0336
95%	1.036	0.9756	0.0241
90%	0.565	1.0000	0.0000

The table gives the central empirical answer. The observed maximum and upper-percentile statistics look large when read fund by fund, but the generated bootstrap p -values show whether they are large relative to the right tail produced by the joint zero-alpha null. The right-tail statistics are not borderline failures to reject; they are far from conventional rejection thresholds. Within the audited WRDS-derived panel, the Carhart benchmark, and the implemented bootstrap design, the best observed funds are statistically indistinguishable from the best funds that luck alone generates under the joint zero-alpha null.

The Monte Carlo standard errors reinforce the same point. They are below one percentage point for the reported statistics, while the p -values are nowhere near 0.05 or 0.10. The result

therefore does not depend on simulation noise at the decision boundary.

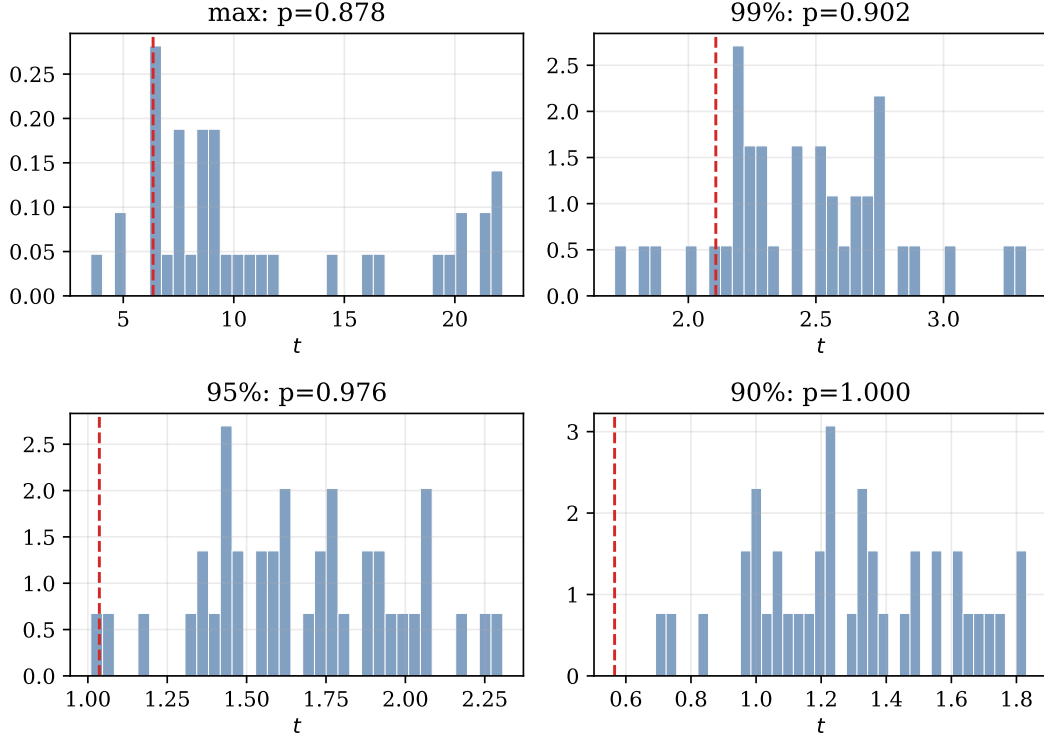


Figure 2: Selected cross-sectional bootstrap null distributions. Red reference lines mark observed order statistics.

Figure 2 makes the table auditable. The red lines mark the observed statistics, while the grey distributions show the no-alpha bootstrap outcomes. For the maximum and upper-tail percentiles, the observed values sit well inside the bootstrap reference distribution rather than in its rejection tail. This is the visual version of the main finding: in a cross-section this large, luck alone often generates very strong-looking winners.

The result should be read narrowly but not weakly. It is evidence about this audited WRDS-derived matrix under the Carhart model and the stated bootstrap design. It is not evidence that no manager can ever have skill, and it does not remove the joint-hypothesis problem from asset pricing. It does say that this particular right tail does not survive the discipline needed to turn impressive fund-level t -statistics into a cross-sectional skill claim.

6 Robustness and Sensitivity

Robustness checks separate implementation choices from economic interpretation. The first concern is studentization. A fund can look more or less exceptional depending on how its alpha standard error is estimated. To study that concern, Table 4 recomputes the observed order statistics using OLS, HC3, and Newey–West standard errors.

Table 4: Observed order statistics under alternative alpha standard errors.

Covariance	Funds	Max	99%	95%
HC3	6,764	6.511	1.999	0.964
NW(12)	6,764	4.528	2.003	1.010
NW(3)	6,764	5.732	2.013	1.008
NW(6)	6,764	4.967	1.979	1.017
OLS	6,764	6.350	2.108	1.036

The maximum statistic remains large under every covariance rule, while the 99th and 95th percentiles stay in the same broad range. This tells us that the visual existence of a right tail is not created by one standard-error convention. It does not by itself establish skill; it simply says the next question still has to be answered with a joint-null bootstrap.

The next check asks whether the main conclusion is an artifact of the 60-month baseline filter. To verify this, Table 5 reruns the same cross-sectional residual bootstrap with looser and stricter fund-history requirements. The economic issue is survivorship. A strict filter can remove short-lived funds, while a loose filter can admit noisier alpha estimates. A robust conclusion should not depend on choosing one of these filters after seeing the result.

Table 5: Filter sensitivity: cross-sectional bootstrap p -values.

Filter	Min months	Funds	Max t	Max p	99% p	95% p
Baseline 60m	60	6,764	6.350	0.8780	0.9024	0.9756
Loose 36m	36	7,899	6.350	0.9615	0.8846	1.0000
Strict 120m	120	4,714	4.386	0.8462	0.6538	0.9615

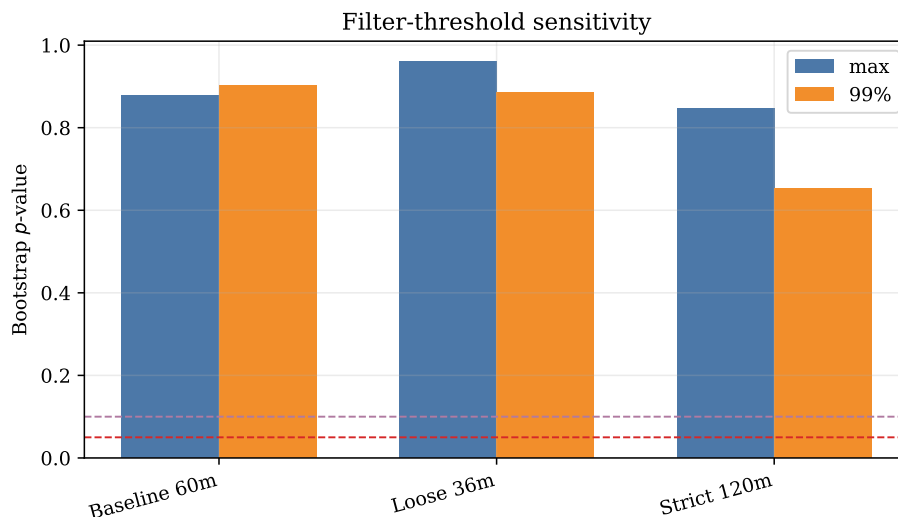


Figure 3: Filter-threshold sensitivity for maximum and 99th-percentile bootstrap p -values.

The filter table should be read as a research-design diagnostic. If the strict rule produced significance while the loose rule did not, the paper would have to frame the evidence as filter-sensitive rather than as broad skill evidence. Figure 3 makes this comparison visible by placing

the same upper-tail statistics on one scale.

The factor-specification check asks a different question: are the apparent winners skilled, or are they earning returns that the baseline factor model fails to absorb? To verify this, Table 6 keeps the baseline 60-month fund universe fixed and reruns the bootstrap under the Carhart model, a three-factor model without momentum, a specification without HML, and a CAPM-only appendix stress test.

Table 6: Factor-specification sensitivity on the baseline fund universe.

Model	Factors	Funds	Max t	Max p	99% p	95% p
CAPM only	Mkt-RF	6,764	6.658	0.6923	0.7692	0.8462
Carhart 4F	Mkt-RF, SMB, HML, UMD	6,764	6.350	0.8780	0.9024	0.9756
FF3 / no UMD	Mkt-RF, SMB, HML	6,764	6.525	0.9231	0.7692	1.0000
No HML	Mkt-RF, SMB, UMD	6,764	6.375	0.8077	0.8462	0.9231

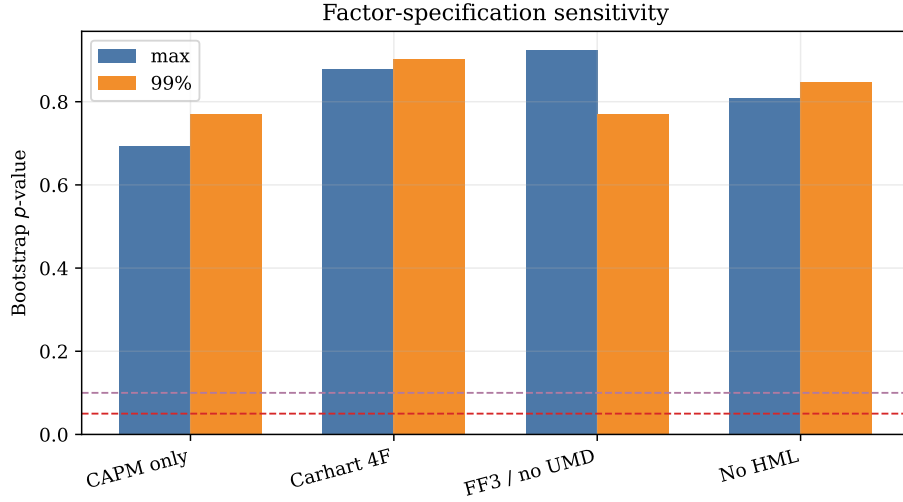


Figure 4: Factor-specification sensitivity for maximum and 99th-percentile bootstrap p -values.

This is the most direct omitted-factor test in the paper. If dropping a factor flips the inference, the right interpretation is not that the omitted-factor model is better. It is that alpha is model-dependent, and that any economic claim must be tied to the benchmark used to define abnormal performance.

Table 7 asks whether the bootstrap conclusion depends on implementation choices after the extreme-return gate has been applied. The primary column is the clean-matrix cross-sectional residual bootstrap with standard degrees of freedom. The unique-dof column retains the older nonstandard degrees-of-freedom rule as a sensitivity check. The block column asks whether a simple one-year moving-block resampling changes the conclusion.

Table 7: Bootstrap robustness: p -values across resampling and data choices.

Statistic	Primary CS	Unique-dof CS	Block CS
max	0.8780	0.6923	0.9615
99.5%	0.9024	0.3077	0.8462
99%	0.9024	0.3077	0.8077
98%	0.9512	0.6538	0.9231
97%	0.9512	0.6923	0.9615
95%	0.9756	0.9231	1.0000
90%	1.0000	1.0000	1.0000

The main message is stable. The primary and block-bootstrap columns both produce high p -values for the maximum and upper percentiles. The unique-dof column is lower for the 99.5% and 99% percentiles, which is informative because it shows that the older degrees-of-freedom rule can materially change tail calibration. It still does not turn the evidence into a strong right-tail rejection, and that is why the standard degrees-of-freedom rule remains the baseline.

The wild bootstrap checks a different failure mode. Residual resampling asks what happens when calendar residual rows are resampled. Wild bootstrapping asks whether the conclusion changes when residual magnitudes are preserved but signs or multiplier weights generate the no-alpha world. Table 8 reports the maximum statistic under standard Mammen weights, time-clustered Mammen weights, and Rademacher weights.

Table 8: Wild-bootstrap robustness for the maximum statistic.

Wild bootstrap	Max t	Max p -value	MC s.e.
mammen independent	6.350	0.6585	0.0741
mammen time clustered	6.350	0.6585	0.0741
rademacher independent	6.350	0.6098	0.0762

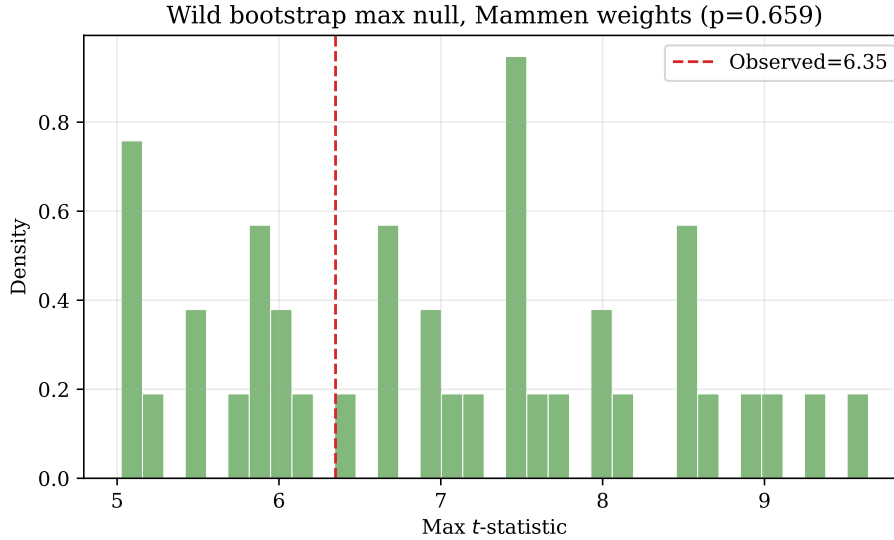


Figure 5: Wild-bootstrap null distribution for the maximum statistic using standard Mammen weights.

The wild-bootstrap p -values are lower than the primary residual-bootstrap p -value, but they remain far above conventional rejection levels. Figure 5 shows why: the observed maximum is large, but it remains a plausible draw from the wild-bootstrap null. This check is useful because it prevents the paper from relying on one resampling design, while also avoiding the false claim that independent fund-time multipliers preserve all cross-sectional dependence.

The full-sample wild bootstrap extends this logic to the unfiltered clean panel. To verify whether the 60-month baseline excludes economically important short-lived funds, Table 9 reruns the wild bootstrap on every clean fund with enough observations for a regression. The table also reports how many fund-level bootstrap p -values fall below 0.05 compared with the count expected under a uniform null.

Table 9: Full-clean-sample wild bootstrap.

Full-sample wild run	Funds	Max t	Max p	MC s.e.	Fund $p < 0.05$	Uniform exp.
Mammen independent	9,263	6.350	1.0000	0.0000	1	463.2
Mammen time-clustered	9,263	6.350	0.9512	0.0336	0	463.2
Rademacher independent	9,263	6.350	0.9512	0.0336	1	463.2

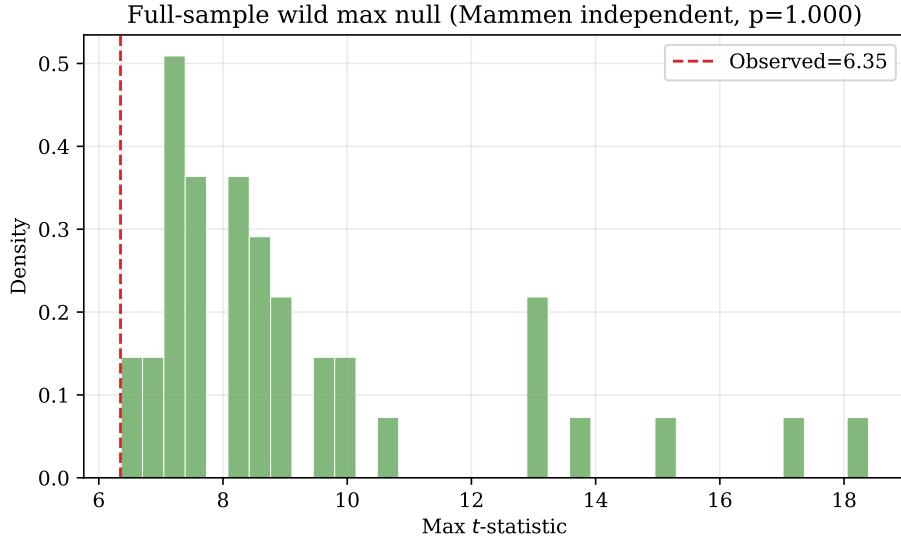


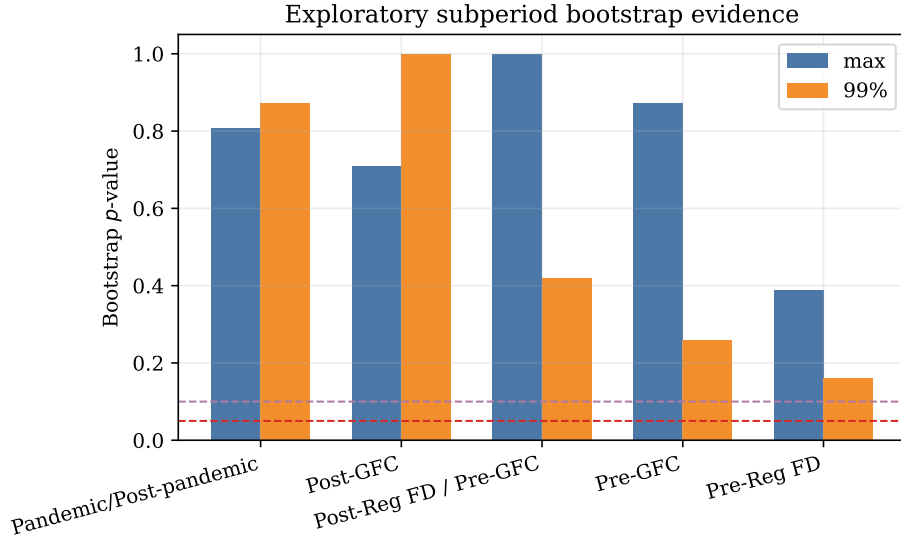
Figure 6: Full-clean-sample wild-bootstrap null distribution for the maximum statistic.

This check matters because short-lived funds can create noisy extreme statistics, while a correct joint-null benchmark also creates noisy extreme statistics. The question is not whether the full sample has larger raw maxima; it is whether those maxima remain unusual relative to a null distribution calibrated to the same full sample.

The subperiod analysis asks a final, more tentative question. If the full sample does not reject the joint null, are there windows where the right tail looks stronger? To study that possibility, Table 10 repeats the bootstrap in pre-Reg FD, post-Reg FD/pre-GFC, pre-GFC, post-GFC, and 2020–2025 pandemic/post-pandemic windows.

Table 10: Exploratory subperiod bootstrap evidence.

Period	Dates	Months	Funds	Max t	Raw max p	Holm max p
Pandemic/Post-pandemic	2020-01–2025-12	72	4,428	3.421	0.8065	1.0000
Post-GFC	2010-01–2025-12	192	5,353	6.350	0.7097	1.0000
Post-Reg FD / Pre-GFC	2000-10–2007-12	87	3,080	3.741	1.0000	1.0000
Pre-GFC	1987-01–2007-12	252	2,548	4.572	0.8710	1.0000
Pre-Reg FD	1987-01–2000-09	165	1,047	4.138	0.3871	1.0000

Figure 7: Exploratory subperiod p -values for maximum and 99th-percentile statistics.

The subperiod evidence does not change the paper’s conclusion. The maximum-statistic p -values are evaluated within each pre-specified window, and the right-tail evidence is not treated as robust unless it survives both the window-specific bootstrap and the multiplicity adjustment. Figure 7 is included to make that comparison visible. It also clarifies why the paper avoids claims such as “skill disappeared after the GFC.” The data can support exploratory subperiod descriptions, but not causal regime statements.

Table 11 reports the sharper version of the same question. For each window it compares the observed maximum with the average and upper tail of that window’s own bootstrap null, then reports raw and Holm-adjusted p -values. This prevents the paper from treating a visually stronger pre-period right tail as evidence of skill unless it is also strong relative to the luck distribution for that exact period.

Table 11: Exploratory subperiod contrasts with multiplicity adjustment.

Period	Dates	Max excess	Null 95% max	Raw max p	Holm max p	Borderline
Pandemic/Post-pandemic	2020-01–2025-12	-4.126	16.632	0.8065	1.0000	No
Post-GFC	2010-01–2025-12	-6.786	23.128	0.7097	1.0000	No
Post-Reg FD / Pre-GFC	2000-10–2007-12	-11.839	23.855	1.0000	1.0000	No
Pre-GFC	1987-01–2007-12	-6.070	17.823	0.8710	1.0000	No
Pre-Reg FD	1987-01–2000-09	0.250	5.958	0.3871	1.0000	No

The anti-insider-trading-law split is not run as a formal test. The panel begins in 1987, so the available pre-ITSFEA window is too short to support credible four-factor bootstrap inference. The paper records that feasibility limit rather than turning a short and underpowered historical slice into an empirical claim.

7 Limitations

The first limitation is identifier coverage. The return and factor files are WRDS-derived inputs, but the analysis matrix does not bundle recoverable fund identifiers, portfolio identifiers, or share-class links alongside the return columns. The data screen can flag extreme returns and reconstruct the month sequence, but the arrays remain anonymous at the column level. The results therefore support cross-sectional right-tail inference, not named-fund attribution.

The second limitation is model dependence. Alpha is defined relative to the Carhart four-factor model. A different benchmark can reclassify what looks like selection skill as factor exposure, or reclassify factor exposure as residual performance. The empirical result should therefore be read as a statement about benchmark-adjusted performance under this model, not as a general theorem about managerial talent.

The third limitation is dependence. The cross-sectional residual bootstrap preserves common calendar draws and uses the residual panel produced by the factor regressions, but it is not a universal solution to every form of serial or cross-sectional dependence. The wild-bootstrap variants answer related but different questions. Independent multipliers are useful for heteroskedasticity checks; time-clustered multipliers speak more directly to common shocks. None of these designs turns an observational mutual-fund panel into an experiment.

The fourth limitation is multiplicity outside the main joint null. Filter thresholds, covariance estimators, subperiod windows, outlier rules, and resampling designs are research choices. The analysis reports Monte Carlo error and treats subperiod results as exploratory. This discipline matters most when a result is near 0.05 or 0.10, where simulation noise or window selection can otherwise carry too much economic interpretation.

The final limitation is reproducibility scope. The reported tables and figures are generated from machine-readable results, and the workflow records the runtime, seeds, generated artifacts, and known limitations. This does not remove the need for judgment in model choice, but it makes the empirical path from data screen to bootstrap inference explicit.

8 Conclusion

This paper studies whether the right tail of mutual fund alpha t -statistics is stronger than luck would produce in a large cross-section. The empirical design combines sample screening, Carhart four-factor regressions, order-statistic inference, and bootstrap null distributions that impose zero alpha fund by fund.

The empirical answer is conservative but strong within its scope. The audited sample contains eye-catching winners when funds are viewed one by one, but the joint-null bootstrap shows that such winners are not unusual in a universe of thousands of funds. The main p -

values are far from conventional rejection thresholds, wild-bootstrap variants do not overturn that conclusion, and subperiod windows do not provide robust contrary evidence. The paper therefore rejects the narrative that the observed right-tail funds are demonstrably better than random luck after the benchmark, data-quality gate, and multiple-testing problem are handled explicitly.

The conclusion is not that mutual-fund skill is impossible. It is that this audited WRDS CRSP-derived panel makes the apparent winners statistically indistinguishable from luck under the Carhart four-factor joint-null design. That finding replaces a fund-level winner narrative with a cross-sectional benchmark for deciding when a mutual-fund performance claim has actually survived the luck problem.

A Algorithmic Details

Data audit. The audit exists to prevent data errors from becoming alpha evidence. It loads the WRDS-derived return and factor matrices, reconstructs the month index from the meta-data summary, flags monthly excess returns below -120% or above 100% , records the flagged observations, and sets those entries to missing in the primary matrix.

Cross-sectional residual bootstrap. This is the primary test because it recreates the cross-section under the joint zero-alpha null. Estimate the factor model fund by fund. Recenter residuals within each fund. For each bootstrap draw, sample common month indices with replacement, pair the sampled factor rows and residual rows, construct no-alpha pseudo-returns, re-estimate all fund regressions, and store the selected order statistics.

Wild bootstrap. The wild bootstrap checks whether the conclusion depends on residual resampling. Estimate the factor model on the original calendar. Recenter residuals. Draw either independent fund-time weights or common time weights, multiply residuals by those weights, add fitted factor returns under zero alpha, re-estimate, and store order statistics.

Verification. The replication workflow regenerates tables and figures from the results file, compiles the manuscript, and checks that the clean matrix contains no unreviewed extreme returns. It also checks that required figures are present and that the compiled document has no unresolved references or citations.

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